Cognitive Models, Crowd Wisdom, Citizen Science, and Frogs: An Extension of Brown et al. (2024)

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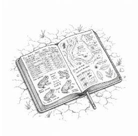
Detecting Frogs

- People take the role of "citizen scientists" by listening to audio clips trying to detect the presence of seven types of frogs by their calls

environment

person

data



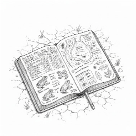
Detecting Frogs

- The same person can listen to multiple clips

environment

person

data



Detecting Frogs

- Multiple people can independently listen to the same clip

environment

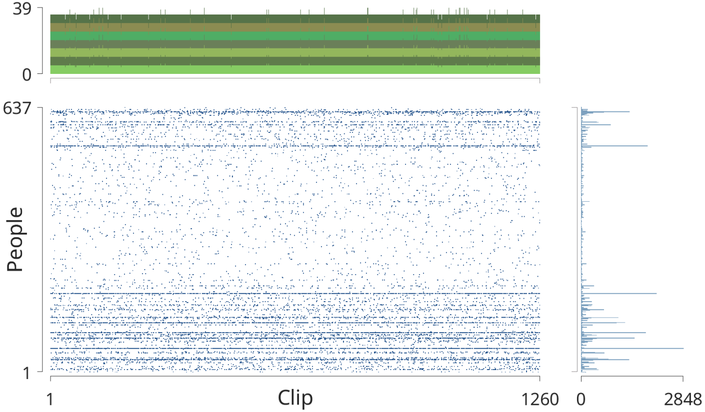
person

data

Behavioral Data

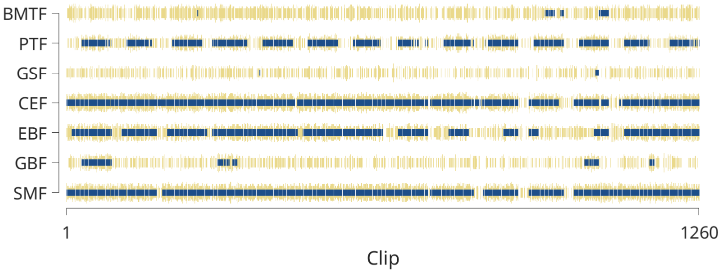
Data

- The joint and marginal distributions of people judging clips shows
  - lots of people judge few clips, but a few people judge many
  - the data collection system aims to have five judgments for each type of frog in each clip



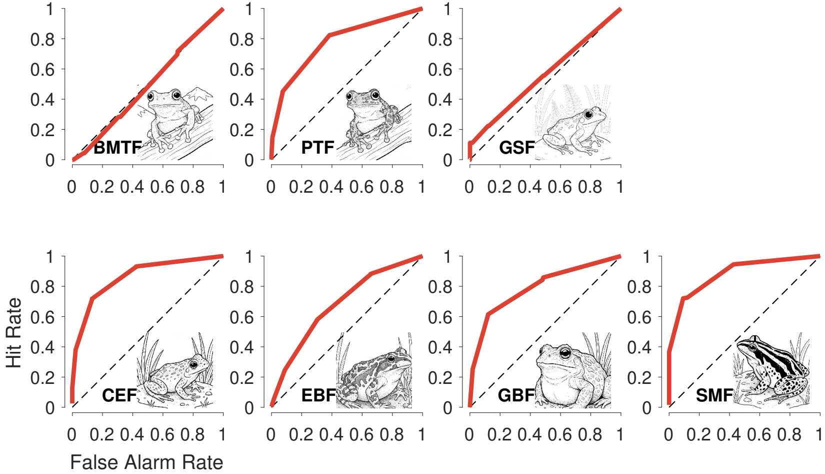
Crowd Proportions and the Truth

- The blue marks indicate the ground truth (expert derived) presence of each type of frog across the sequence of clips
- The yellow bars show the proportion of people who judged a frog was present in each clip



Crowd Performance

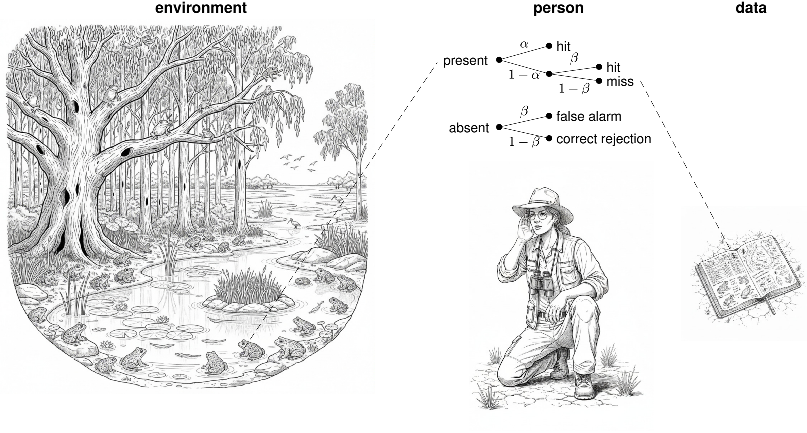
- ROC curves show reasonable wisdom the crowd performance for five of the frog types
  - the BMTF and GSF tree frogs are not well detected
  - both are rarely present but people often report detecting them



Model-Based Wisdom of the Crowd

Model-Based Wisdom of the Crowd

- The model-based approach uses cognitive models to explain how the behavior data were produced (Brown et al., 2024; Lee, 2024)
  - make inferences about the cognitive abilities and biases of individuals
  - make inferences about latent states of nature



One-High Threshold Model

- The model assumes frog f in clip j is either $\tau_{fj} = 1$ present or $\tau_{fj} = 0$ absent with a base-rate that depends on the frog ϕ_f

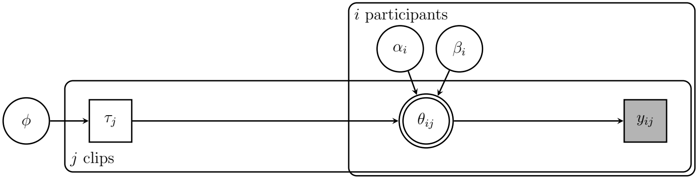
$$\begin{aligned} \tau_{fj} &\sim \text{Bernoulli}(\phi_j) \\ \phi_j &\sim \text{uniform}(0,1) \end{aligned}$$

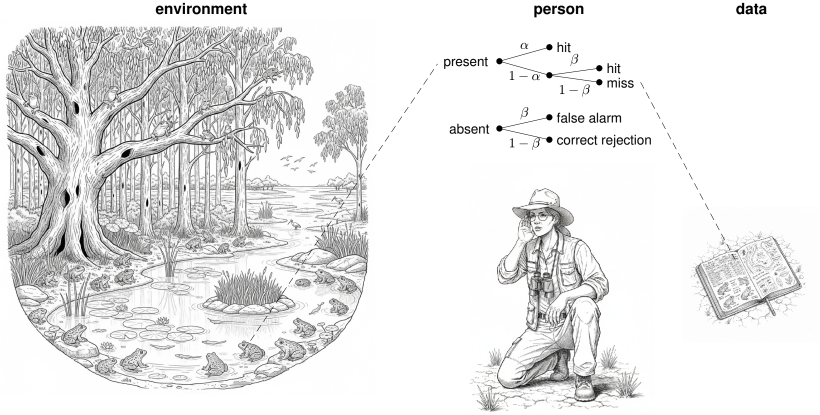
- Individual i judges frog f in clip j to be $y_{ijf} = 1$ present with probability θ_{ijf} given by the one-high threshold model

$$\begin{aligned} \alpha_i, \beta_i &\sim \text{uniform}(0,1) \\ \theta_{ijf} &= \begin{cases} \alpha_i + (1 - \alpha_i)\,\beta_i & \text{if } \tau_{fj} = 1 \\ \beta_i & \text{if } \tau_{fj} = 0 \end{cases} \\ y_{ijf} &\sim \text{Bernoulli}(\theta_{ijf}) \end{aligned}$$

- Implemented the model in JAGS and Stan

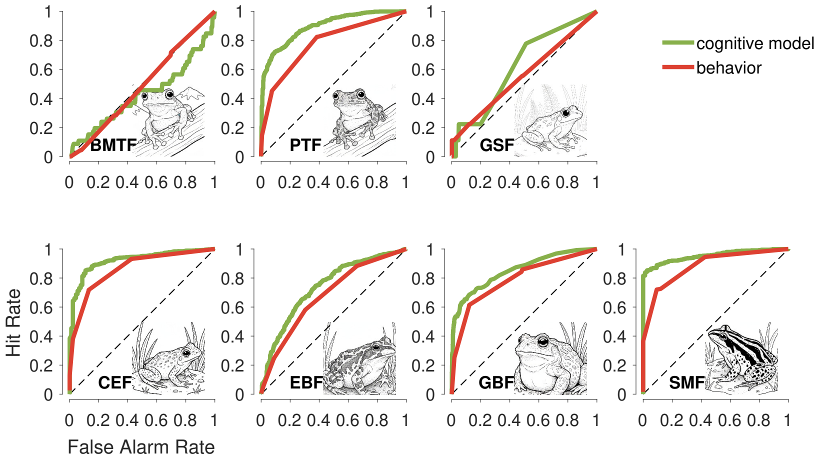
Inferences about both the Stimulus and Cognitive Processes





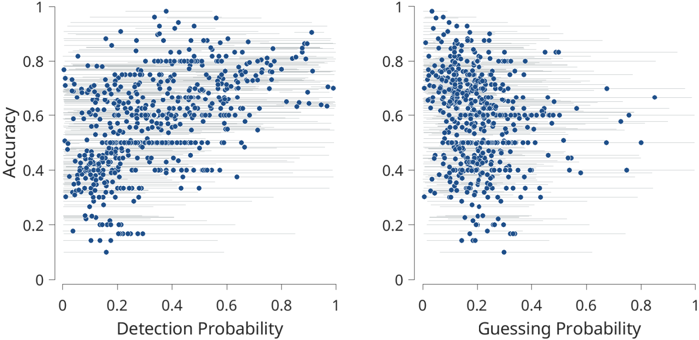
Model-Based Crowd Performance

- The model-based approach performs better than the standard wisdom of the crowd aggregation of behavioral judgments



Inferences about Detection and Guessing Bias

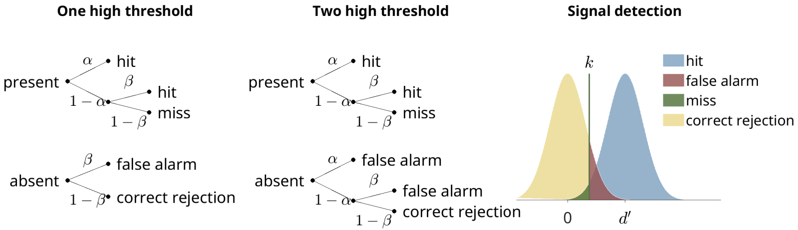
- Inferences about detection ability α_i and guessing bias β_i provide predictions about the performance of individual i
  - the accuracy of these predictions allows, for example, "yes" judgments from non-experts to be explained away as guessing



Which Cognitive Model?

Three Cognitive Models

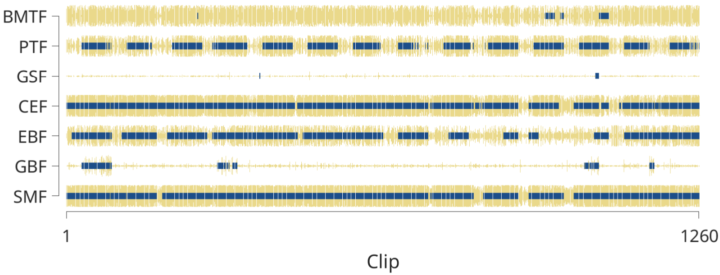
- There are many other possible cognitive models of the detection decisions
  - e.g., the two high threshold model, signal detection models, ...
- They all seem to perform very similarly
  - same ROC curves, and an "ability" parameter that relates to observed accuracy



Modeling the Environment

Model-Based Crowd and the Truth

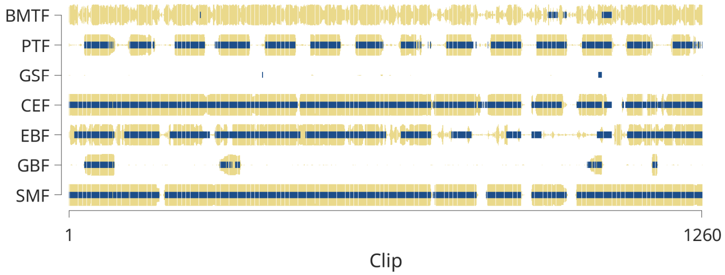
- Brown et al. (2024) point out that their model-based approach assumes independence over both scientists and clips
  - reasonable assumption for scientists, but there are obvious environmental dynamics that create dependencies over clips
  - "frogs come and frogs go" . . .



Environment Dynamics Model and the Truth

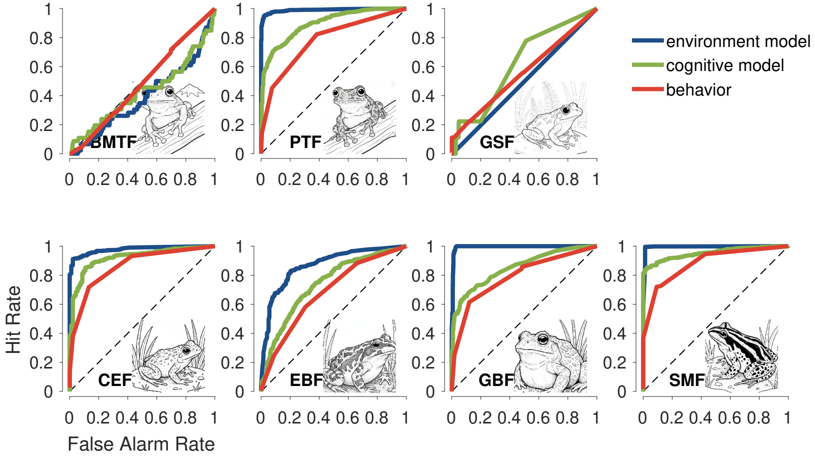
- The environmental model has a switching probability γ_f for frog f

$$
\begin{aligned}
\gamma_f &\sim \mathrm{uniform}(0,1) \\
\tau_{f1} &\sim \mathrm{Bernoulli}(\frac{1}{2}) \\
\tau_{fk} &\sim \begin{cases} \mathrm{Bernoulli}(\gamma_f) & \text{if } \tau_{f(k-1)} = 1 \\ \mathrm{Bernoulli}(1-\gamma_f) & \text{if } \tau_{f(k-1)} = 0,\ k > 1 \end{cases}
\end{aligned}
$$



Environment Dynamics Model Performance

- Performance is now near-perfect for three of the frog types, and better for the other two with behavioral signal



Discussion

Considering Environmental Structure

- It is easy as a cognitive modeler to focus too hard on internal cognitive processes
- The dynamics of the environment also influence observed behavior, and should be part of a complete model of the data-generating process

- Focusing on environmental regularities for this frog task could lead to improvement in
  - model development: more complicated environmental models are possible
    - e.g., model the covariances in the presence of frog types
  - data collection: presenting people with random clips corrupts the environmental structure their cognition may be able to use to make better judgments (Whitney, Sweeny, & Corbett, 2014)

Thanks!

References

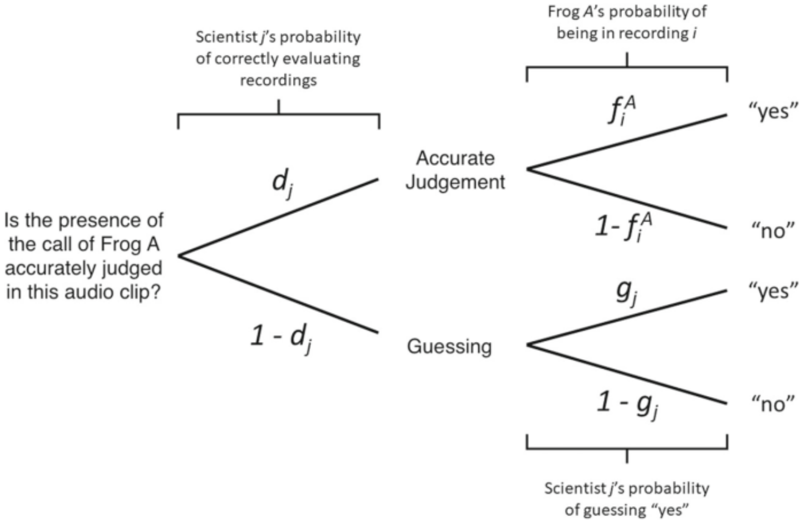
Brown, S. D., Sumnall, J., & Smith, P. R. (2024). Using a cognitive model to understand crowdsourced data from citizen scientists. *Behavior Research Methods*, *56*(1), 421–434. doi: 10.3758/s13428-023-02289-w

Lee, M. D. (2024). Using cognitive models to improve the wisdom of the crowd. *Current Directions in Psychological Science*, *33*(5), 444–451. doi: 10.1177/09637214241264292

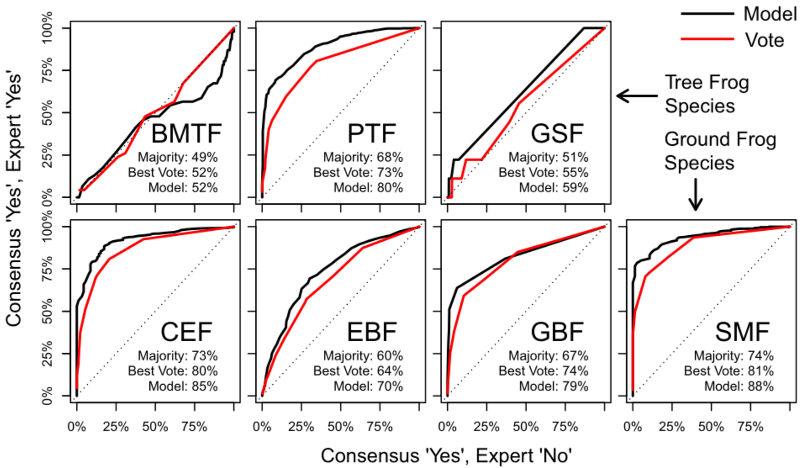
Whitney, D., Sweeny, T. D., & Corbett, J. E. (2014). Visual ensemble statistics are represented as gradients of crowding. *Nature Neuroscience*, *17*(1), 12–14. doi: 10.1038/nn.3603

Brown et al. (2024) Model and Results

Brown et al. (2024) Model

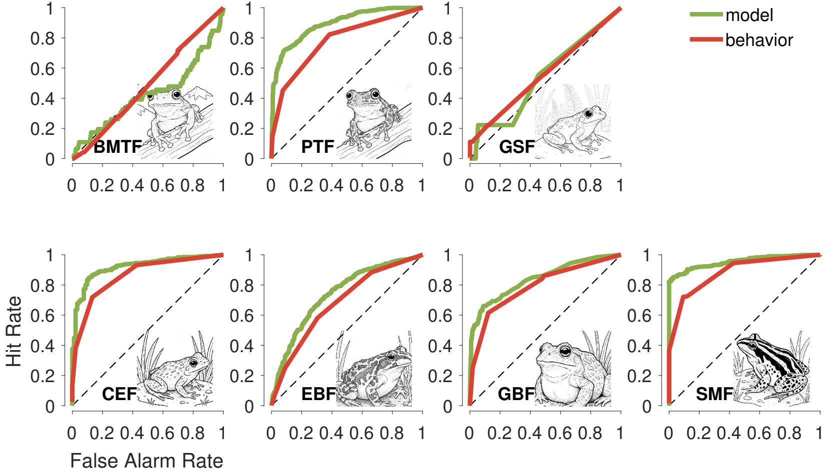


Brown et al. (2024) Results

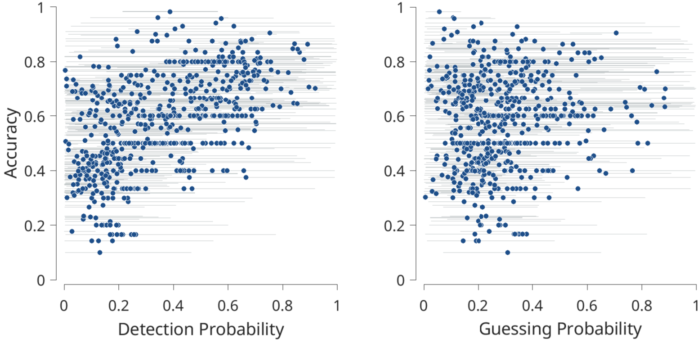


Performance of Other Cognitive Models

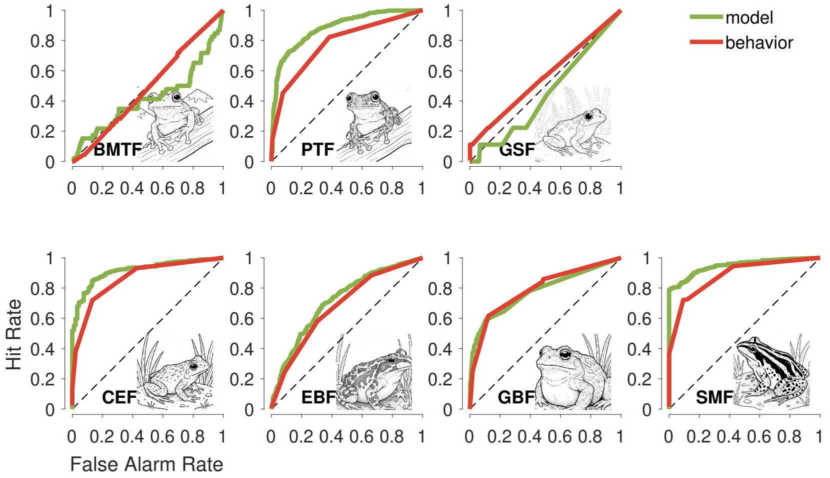
Two High Threshold Crowd Performance



Two High Threshold Parameter Inferences



Signal Detection Crowd Performance



Signal Detection Parameter Inferences